Date: 22.8.2023

Tuesday.

TOC 21IS44

CIE-2

Scheme & Solution

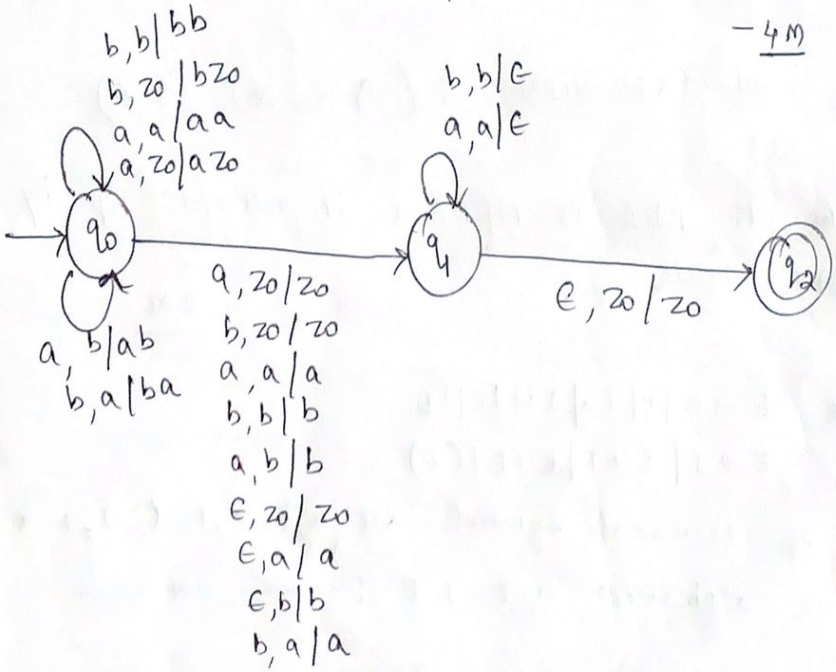
1) PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ — 2M

States, I/P, ↓ stack symbols; q_0 initial state; Z_0 stack top symbol; F set of final states

δ is a transition fn. which takes I/P in the form of a triple (q, a, X)
state, I/P, stack symbol.

Def^n of Instantaneous Description - 1M

PDA to recognize all palindromes over $\Sigma = \{a, b\}$. — 4M



Trace for abba - 1M

1b] $S \to aaS \mid bbS \mid Saa \mid Sbb \mid abSab \mid abSba \mid baSba \mid baSab \mid \epsilon$

$L = \{ w \in \{a,b\}^* \mid w$ contains even no of a's & even no of b's $\}$.

2a] Steps:

① Starting symbol $\delta(q_0, \epsilon, Z_0) = (q, SZ_0)$

② For each variable A

$\delta(q, \epsilon, A) = \{ (q, \beta) \mid A \to \beta$ is a production of $P \}$

③ For each terminal a, $\delta(q, a, a) = \{(q, \epsilon)\}$

④ Final transition: $\delta(q, \epsilon, Z_0) = (q, \epsilon)$

CFG to PDA is NPDA & it accepts i/p by empty stack

2M

* The $I \to a \mid b \mid Ia \mid Ib \mid I0 \mid I1$

$E \to I \mid E*E \mid E+E \mid (E)$

Set of terminal symbols are $\{a, b, 0, 1, (,), +, *\}$

stack alphabets are I, E & ——— " ———

① $\delta(q, \epsilon, Z_0) = (q, IZ_0)$

② $\delta(q, \epsilon, I) = \{(q, a), (q, b), (q, Ia), (q, Ib), (q, I0), (q, I1)\}$

③ $\delta(q, \epsilon, E) = \{(q, I), (q, E*E), (q, E+E), (q, (E))\}$

$\delta(q, a, a) = \{(q, \epsilon)\}$ $\delta(q, (, ()) = \{(q, \epsilon)\}$

$\delta(q, b, b) = \{(q, \epsilon)\}$ $\delta(q,),)) = \{(q, \epsilon)\}$

$\delta(q, 0, 0) = \{(q, \epsilon)\}$ $\delta(q, +, +) = \{(q, \epsilon)\}$

$\delta(q, 1, 1) = \{(q, \epsilon)\}$ $\delta(q, *, *) = \{(q, \epsilon)\}$.

(5) $\delta(q, \epsilon, Z_0) = (q, \epsilon)$

3M

2b] Languages accepted by PDA are:

(1) Final State . 2 * 5 * 2 = 5M

(2) Empty Stack

1.5 for description & 1 M for example.

3.a] Steps to find an equivalent CFG from a given PDA : 2M

Given PDA $P = (\{p, q\}, \{0, 1\}, \{X, Z\}, \delta, p, Z, p\}$

where δ is given by

(1) S productions are given by $S \to [q_0 Z q]$ for every $q \in Q$.

(2) Each erasing move $\delta(q, a, Z) = (q', A)$ induces production $[qZq'] \to a$

(3) Each non-erasing move $\delta(q, a, Z) = (q_1, Z_1, Z_2 \ldots Z_m)$ induces many productions of the form

$[qZq'] \to a[q_1 Z_1 q_2][q_2 Z_2 q_3] \ldots [q_m Z_m q']$ where each state $q', q_2 \ldots q_m$ can be any state in Q.

$S \to [p, z, q]$

$[p, z, q] \to 0[p, x, q][q, z, q]$

$[q, z, q] \to 0[p, x, q][q, x, q]$

$[p, x, q] \to 1$

$[q, z, q] \to \epsilon$ 4M

$[q, x, q] \to \epsilon$

$[q, x, q] \to 1$.

3b]

$S \to AB \mid ABC$

$A \to BA \mid BC \mid a \mid \epsilon$

$B \to AC \mid CB \mid b \mid \epsilon$

$C \to BC \mid AB \mid A \mid c$

* Eliminate ϵ- production

i.e $S \to AB \mid ABC \mid B \mid A \mid AC \mid BC \mid C$

$A \to BA \mid BC \mid A \mid B \mid C \mid a$

$B \to AC \mid CB \mid b \mid C$ $\to$ 2M

$C \to BC \mid AB \mid A \mid B \mid C \mid c$

After unit production removal

$S \to ABC \mid AB \mid AC \mid BC \mid BA \mid CB \mid a \mid b \mid c$

$A \to AB \mid AC \mid BA \mid BC \mid CB \mid a \mid b \mid c$

$B \to AC \mid CB \mid BC \mid AB \mid b \mid c$

$C \to AB \mid BC \mid c$ $\to$ 2M

4a] DPDA def^n:

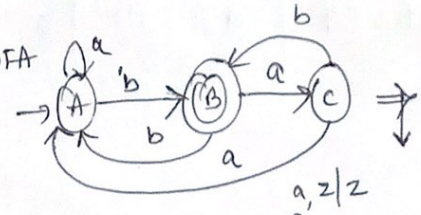
Let the PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ in which

① $\delta(q, a, Z)$ has atmost one entry.

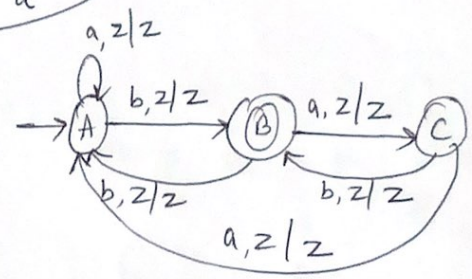
② If $\delta(q, \epsilon, Z)$ is not empty then $\delta(q, a, Z)$ must be empty where $q \in Q$, $a \in \Sigma$ & $Z \in \Gamma$. 2M

δ	a	b
$\rightarrow$ A	A	B
* B	C	A
C	A	B

$\Rightarrow$ DFA



$\Rightarrow$



5a] proof. 5M

5b] $S \rightarrow aB \mid bA$

$A \rightarrow a \mid aS \mid bAA$

$B \rightarrow b \mid bS \mid aBB$

string: aabbabab.

$S \rightarrow aB$

$aaBB$

$aabSB$

$aabbAB$

$aabbabS$

$aabbabaB$

$aabbabab$

$S \rightarrow aB$

$S \rightarrow aaBB$

$S \rightarrow aaB$

4b]. GNF grammar:

$X \to xX, \mid$ ie A non terminal generating a terminal followed by any no of non-terminals.

$A \to BC,\ B \to CA \mid b,\ C \to AB \mid a$

Rename variables $A = A_1,\ B = A_2,\ C = A_3$

$A_1 \to aA_3A_4 \mid aA_3A_4 \mid aA_1A_3 \mid aA_3$

$A_2 \to$